

Physical Chemistry (I)

Lecture 10



1차 중간고사 안내

- 시간: 4월 8일(목) 오후 1:30~2:30
- 장소: 화학관 315호(교수계획표에 나온 수업 장소)
- 계산기 불필요

시험 범위

- 1장 “열역학의 기본 개념들”
- 2장 “기체의 성질”
- 3장 “열역학 제1법칙”
- 4장 “엔트로피”

시험 문제

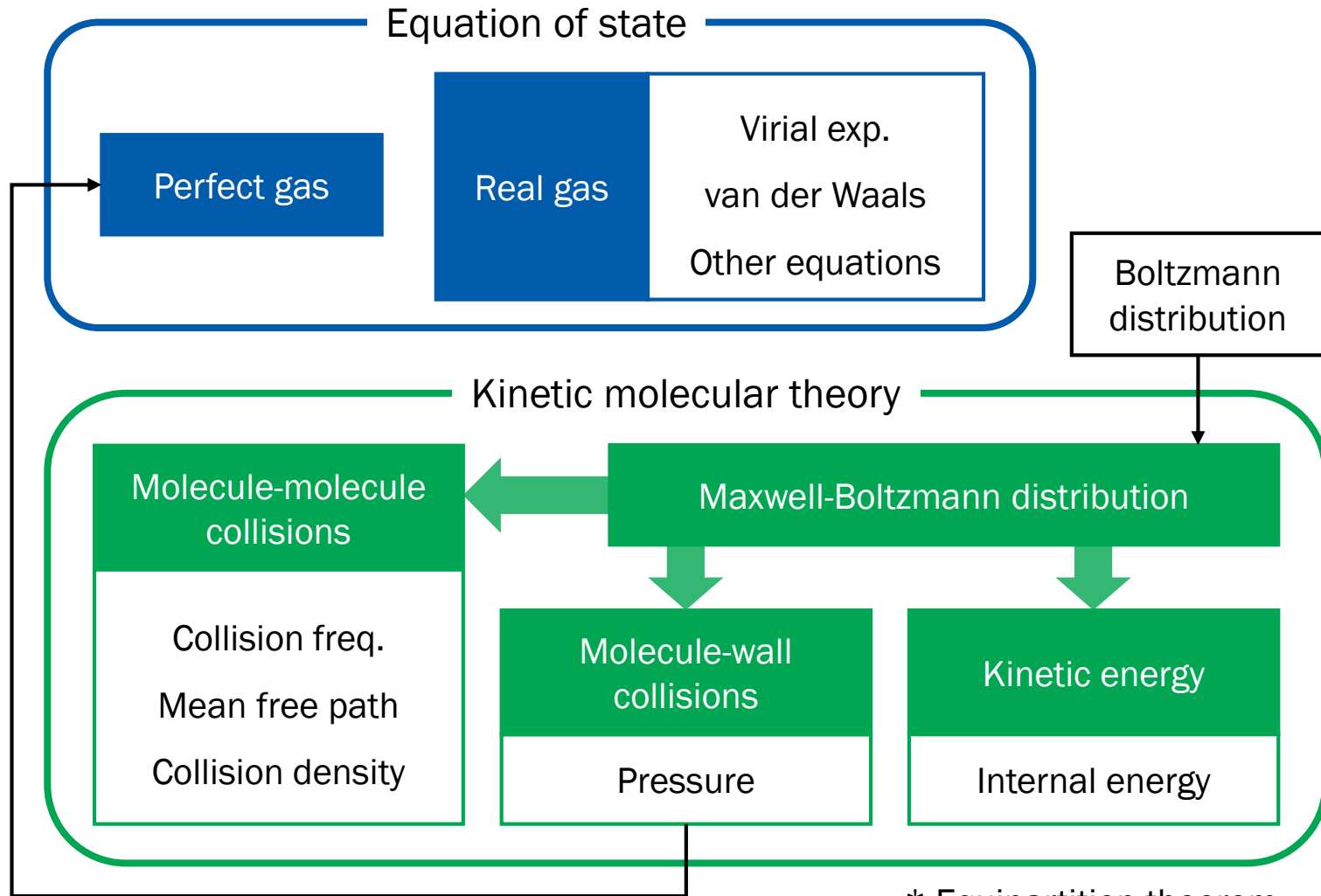
- 총 네 문제
- 과제 문제 혹은 그 변형
- 수업 시간에 배운 공식 유도
- 응용 문제: 개념 + 유도

Concepts in Chpt 1

- System and surroundings
- Classification of systems
- System properties: extensive vs. intensive
- Equilibria: mechanical, diffusive, and thermal
- The zeroth law of thermodynamics
- State of a system
- State change



Chapter 2



* Equipartition theorem

Concepts and Derivations in Chpt 2

- [C] Perfect gas vs. real gas
- [C] Equation of state, perfect gas law
- [C] Compression factor
- [C] Virial expansion
- [C] Boyle temperature
- [C, D] van der Waals equation
- [D] van der Waals and virial
- [C] Other real gas equations: no need to memorize



Concepts and Derivations in Chpt 2

- [C] Kinetic molecular theory
- [C] Boltzmann distribution
- [D] Maxwell-Boltzmann distribution
- [D] Characteristic speeds
- [D] Collision frequency, mean free path, collision density
- [D] Pressure
- [D] Internal energy
- [C] Equipartition theorem



Chapter 3 (1)

First Law of Thermodynamics

$$dU = \delta q + \delta w$$

Work

$$\delta w = -p_{\text{ex}}dV + \delta w_{\text{add}}$$

- Constant- p change
- Reversible change

Heat

$$\delta q = C dT$$

- Constant- V change

$$C_V = (\partial U / \partial T)_V$$

Thermodynamics

Mathematics

State function

Exact differential

Path function

Inexact differential

Differential form of U

$$dU = \left(\frac{\partial U}{\partial V} \right)_T dV + \left(\frac{\partial U}{\partial T} \right)_V dT$$

- Internal pressure π_T
- Response functions α , κ_T

Chapter 3 (2)

Thermodynamics	Mathematics
State function	Exact differential
Path function	Inexact differential

Differential form of U

$$dU = \left(\frac{\partial U}{\partial V}\right)_T dV + \left(\frac{\partial U}{\partial T}\right)_V dT$$

Enthalpy

$$H = U + pV$$

$$dH = \delta q_p$$

Differential form of H

$$dH = \left(\frac{\partial H}{\partial p}\right)_T dp + \left(\frac{\partial H}{\partial T}\right)_p dT$$

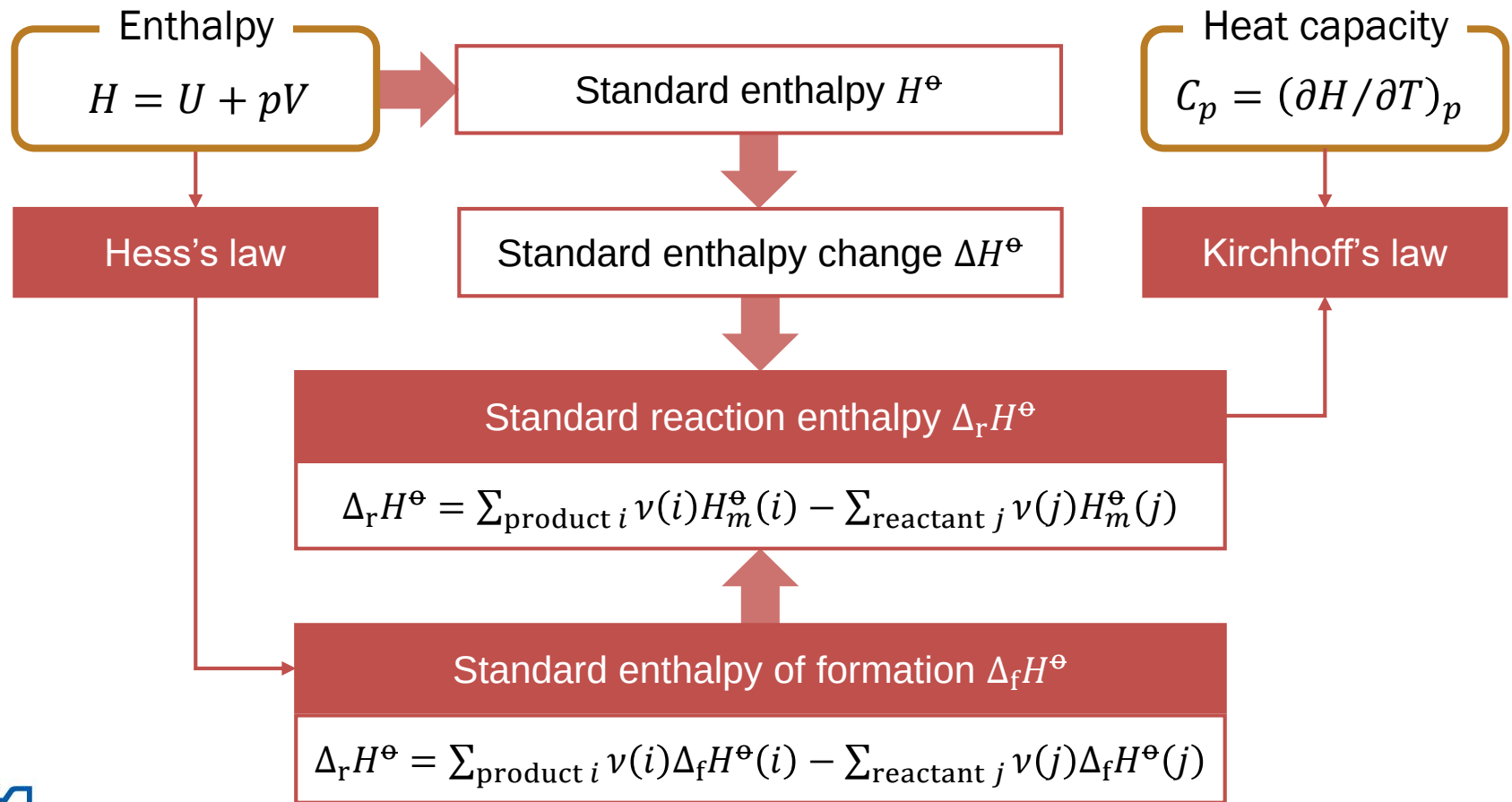
$$C_V = (\partial U / \partial T)_V$$

C_V and C_p

- Perfect gas
- Adiabatic change: $\delta q = 0$

$$C_p = (\partial H / \partial T)_p$$

Chapter 3 (3)



Concepts and Derivations in Chpt 3

- [C] First law of thermodynamics
- [D] Expansion work
 - Isochoric and isobaric change
 - Free expansion
 - Reversible(perfect gas, isothermal) change
- [D] Heat
 - Isochoric and isobaric change
 - Internal energy and isochoric heat capacity



Concepts and Derivations in Chpt 3

- [C] State and path functions
- [C, D] Differential form of internal energy
- [C] Response functions
- [D] Derivation of $(\partial U / \partial T)_p$
- [C, D] Enthalpy
 - Reversible isobaric change
 - Enthalpy and isobaric heat capacity
- [C, D] Differential form of enthalpy

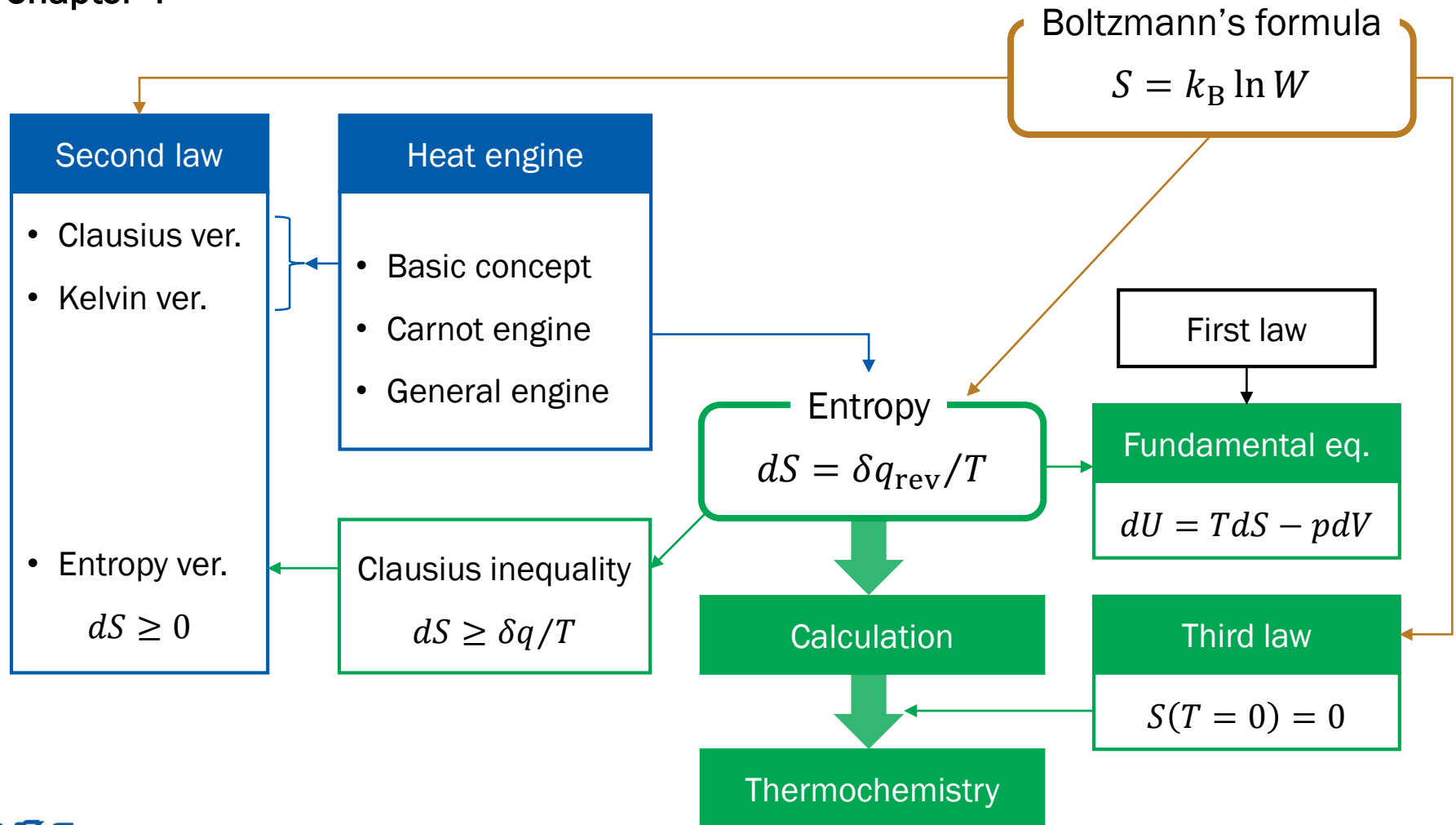


Concepts and Derivations in Chpt 3

- [D] Heat capacities for perfect gas
- [C, D] Adiabatic change
 - Work for perfect gas
 - Reversible, perfect-gas relations: V - T and V - p
- [C] Standard enthalpy
 - Standard reaction enthalpy
 - Standard enthalpy of formation
- [C] Hess's law
- [D] Kirchhoff's law



Chapter 4



Concepts and Derivations in Chpt 4

- [C] Heat engine
- [C, D] The second law of thermodynamics
- [C] Engine efficiency
- [C, D] Carnot cycle and Carnot efficiency
- [C] Generalization of Carnot efficiency
- [C] Entropy
- [D] Entropy calculation
 - Isothermal reversible expansion
 - Entropy change of surroundings



Concepts and Derivations in Chpt 4

- [C, D] Clausius inequality
 - Entropy version of the second law
 - Entropy and spontaneity
- [C] Fundamental equation of thermodynamics
- [D] Entropy calculations
 - Heat transfer
 - Expansion
 - Phase transition
 - Heating



Concepts and Derivations in Chpt 4

- [C] Nernst heat theorem and the third law
- [C] Thermochemistry of entropy
 - Standard entropy
 - Standard reaction entropy
- [C] Entropy as an index of exhaustion
- [C] Boltzmann's formula
- [D] Applications of Boltzmann's formula
- [C] Reinterpretations of the laws
- [C] Qualitative estimation of molecular entropy

